22 Ways to Kill Event Density

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Event Density (ED) is a contrast-first substrate ontology for physics. Its primitive is neither the event nor the relation but contrast (a difference, a gradient, a participation pattern) from which both co-arise, together with commitment (P11), the substrate's irreversible act that converts relational potential into determinate fact. From this base, the framework derives quantum mechanics, gauge structure, gravity, and black-hole thermodynamics as downstream structural consequences, each tiered explicitly by how much of a given result is forced from the primitives and how much is inherited from measurement.

The full corpus, currently more than 125 papers, is public and version-controlled.

Full corpus: https://github.com/allen-proxmire/ED-generative

This document is not an argument for why ED is right. It is the opposite: a list of specific, numeric ways it could be shown wrong. Twenty-two are live bets today, checkable against data that already exists or experiments already running. Two more are real predictions without a clean kill condition built yet. Three have already been checked and passed, listed here so the bar for the rest is clear.

Try to kill it.

Gravity, Dark Matter, Dark Energy

1. MOND transition acceleration. $a_0 = cH_0/(2\pi) \approx 1.08 \times 10^{-10}\,\mathrm{m/s^2}$: the acceleration scale where galaxy rotation curves depart from plain Newtonian gravity, derived here from the cosmic expansion rate rather than fit to galaxy data. *Kill it:* a_0 tracks H_0 wrong as the Hubble tension resolves, or wide binary stars at this acceleration obey plain Newton with no anomaly.

2. Baryonic Tully-Fisher slope. $v^4 = GMa_0$, slope exactly 4, zero intrinsic scatter: the deep-MOND asymptote of the galaxy mass-velocity relation. *Kill it:* any measured slope other than 4, or scatter beyond observational uncertainty, checkable now against the public SPARC catalog.

3. Preferred-frame parameters. $\alpha_2 = 0$ exactly; α_1 suppressed at least 70 orders of magnitude below the experimental bound. *Kill it:* any measured $\alpha_1 \gtrsim 10^{-5}$, or any nonzero α_2, from lunar laser ranging or pulsar timing.

4. Lensing identity. No gravitational slip beyond the standard post-Newtonian order, and no vector field available to patch a mismatch. *Kill it:* a measured slip the metric-only coupling cannot accommodate.

5. Cosmological-constant magnitude. The smallness of Λ is reframed as $(H_0/M_P)^2 \approx 10^{-122}$, not a separate coincidence to explain. *Kill it:* independent evidence that $\rho_\Lambda \neq (3/8\pi)H_0^2 M_P^2$.

6. Scalar gravitational-wave polarization. The extra scalar mode (the "khronon") should appear as a faint polarization component in GW signals, weaker than the tensor modes by construction. Long-term, not a present-day test, but a real structural prediction with a defined kill condition once sensitivity improves.

7. Universal Lorentz violation. Matter and gravity share exactly one causal cone; any Lorentz violation is universal, not species-dependent. *Kill it:* measured differential Lorentz violation between matter species, at the 10^{-20} level current bounds probe.

Black Holes and Information

8. Non-thermal Hawking correction. $n_H(\omega) = n_H^{\text{Planck}}(\omega)[1 - c_1(\omega/\omega_c)^2 + ...]$: a sign-definite, mass-dependent deviation from an exactly thermal Hawking spectrum. *Kill it:* analog-gravity horizons (BEC, fluid) come out exactly thermal, with no deviation of this sign and scaling.

9. Planck-mass remnant. Evaporation halts rather than completing to $M \to 0$. *Kill it:* late-stage evaporation observed to complete, or a remnant found at a mass other than the Planck mass.

10. Page-curve shape. Entanglement entropy rises, turns over, and declines, driven by a finite entanglement-bandwidth cap. *Kill it:* continued monotonic entropy growth with no turnover, or a plateau instead of a decline after turnover.

11. Extremal-horizon pair emission. Extremal-configuration analog horizons should show zero thermal pair-emission. *Kill it:* thermal pair-emission detected in an extremal-configuration analog horizon, across multiple platforms.

Matter Sector: Gauge Groups, Forces, Dimensions

12. No chiral abelian force. The single-channel (abelian) coupling is chirality-blind by construction: there is no way to get a parity-violating $U(1)$ force out of this substrate. *Kill it:* discovery of a chiral (parity-violating) $U(1)$ gauge force.

13. No stable $SU(N{\geq}4)$. Stable gauge groups are capped by an internal channel-stability count; nothing larger than $SU(3)$ should be stable. *Kill it:* discovery of a stable fundamental $SU(N{\geq}4)$ sector.

14. Spin-1/2 and chirality from relationality. Spin-1/2 and net chirality are argued to be forced by channel topology and relational tethering, not merely posited. *Kill it:* showing the substrate is structurally unable to carry net chirality or the spin double cover.

15. Three spatial dimensions from a topological lock. The arrow needs to hold a committed order against being smoothly undone. The claim is that a link (two threaded loops) is the simplest structure that holds this way, and a link only holds in exactly three spatial dimensions. *Kill it:* showing the substrate holds its order without spatial linking, or that the linking mechanism fails specifically in 3D. *(Honest flag: this one has an open premise currently under active test.)*

Quantum Foundations and Soft Matter

16. Quantum-to-classical decoherence wall. A sharp, architecture-independent decoherence boundary in matter-wave interferometry, extrapolated to 140-250 kDa, directly in the territory of groups already pushing large-molecule interferometry to higher masses. *Kill it:* interference succeeding cleanly at 250 kDa or above with no wall, or no sharp wall at any mass.

17. Second-harmonic decoherence signature. Alongside #16: 3-6% second-harmonic content in interference fringe residuals near the quantum-classical crossing. *Kill it:* pure first-harmonic decay observed near the crossing, the standard decoherence prediction.

18. FRAP front-radius exponent. In concentrated protein solutions (BSA), a specific front-radius exponent $\alpha_R = 1/6$ in fluorescence-recovery experiments. *Kill it:* $\alpha_R \geq 0.35$, or a plain Fickian front winning at all tested concentrations.

19. AFM dewetting invariance. A specific cross-scale invariant in atomic-force-microscopy dewetting measurements, median value -1.30. *Kill it:* a measured median outside $[-1.80, -0.90]$.

20. Envelope modulation at bifurcation. A slow envelope modulation at a specific frequency ratio ($\omega_v \approx 8\gamma_{\text{dec}}$), predicted to hold across ten or more orders of magnitude in decoherence rate, from optomechanics to protein-diffusion residuals. *Kill it:* no envelope detected at the predicted frequency band in any of three or more independent datasets.

21. Synchronized cosmic-boundary covariation. Six independent physical projections (from black-hole horizons to galactic acceleration to lab-scale correlation budgets) trace to one boundary and should co-vary as Hubble-tension-class measurements shift. *Kill it:* any two of the six projections moving out of sync.

22. Electrical-circuit consistency test. ED's canonical governing PDE, in its spatially-uniform limit, is exactly isomorphic to a parallel RLC circuit: a capacitor (the density channel) parallel to a resistor (dissipation) and to a series inductor-resistor branch (the participation channel). Four breadboard configurations (pure RC, pure LR, overdamped RLC, underdamped RLC) extract the same four substrate parameters two independent ways. *Kill it:* the parameters extracted from the RC and LR runs, plugged into the RLC prediction with no free adjustment, disagree with the measured RLC waveform by more than component tolerance (about 1%). Benchtop test, under $50 in parts, one afternoon.

Provisional (real predictions, no clean kill condition built yet)

P1. Activity-dependent weak lensing. Dynamically active systems should show enhanced weak lensing versus quiescent systems of the same baryonic mass. The discriminator that would separate this from ordinary baryonic feedback is not yet built.

P2. Gravitational-wave merger lag. A dynamical delay in binary mergers versus the standard-relativistic merger time should exist. Only its existence, not a specific floor value, is argued for so far; it becomes refutation-grade once that floor is derived.

Already Passed (the floor, not the pitch)

- **Factor-of-two light bending** (deflection = 2x Newtonian), ray-traced 2.09 vs. the Newtonian 2.00.
- **Gravitational-wave speed equal to** c, structurally exact, consistent with GW170817 to 1 part in 10^{15}.
- **Universal Degenerate-Mobility law** for concentrated protein solutions, $D(c) = D_0(1 - c/c_{\mathrm{max}})^{\beta}$, validated across 10 materials, $R^2 > 0.986$, published May 2026.

If these had gone wrong, none of the above would exist.